

Programs on Problem Solving**a: Solve the 8 Queens Problem****Python code:**

```
N = 8
```

```
def print_solution(board):
```

```
    for row in board:
```

```
        print(" ".join("Q" if col else "." for col in row))
```

```
    print("\n")
```

```
def is_safe(board, row, col):
```

```
    for i in range(row):
```

```
        if board[i][col]:
```

```
            return False
```

```
    if col - (row - i) >= 0 and board[i][col - (row - i)]:
```

```
        return False
```

```
    if col + (row - i) < N and board[i][col + (row - i)]:
```

```
        return False
```

```
    return True
```

```
def solve_n_queens(board, row):
```

```
    if row == N:
```

```
        print_solution(board)
```

```
        return True
```

```
    found = False
```

```
    for col in range(N):
```

```
        if is_safe(board, row, col):
```

```
            board[row][col] = 1
```

```

        found = solve_n_queens(board, row + 1) or found
        board[row][col] = 0

    return found

board = [[0] * N for _ in range(N)]
solve_n_queens(board, 0)

```

output:

```

Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . Q . . .
. . Q . . . . .
. . . . . Q . .
. Q . . . . . .
. . . Q . . . .

```

Result:

Thus the program of placing 8 queens on an 8x8 chessboard was executed successfully.